SESSION 2 — Types of Data & Measurement Scales

Task 1

Question:
Given the following columns from a mock Zomato restaurant dataset — Restaurant Name, Cuisine Type, Average Cost for Two, User Rating, City, and Open/Closed Status — classify each column as either Numerical (Continuous or Discrete) or Categorical (Nominal or Ordinal).

Answer:

Column	Data Type	Scale	Explanation
Restaurant Name	Categorical	Nominal	It is a name used to identify a restaurant and has no natural order.
Cuisine Type	Categorical	Nominal	Cuisine types such as Indian, Chinese, and Italian are categories without any ranking.
Average Cost for Two	Numerical – Continuous	Ratio	Cost is numerical, can have decimal values, and has a meaningful zero.
User Rating	Numerical – Continuous	Ordinal	Ratings have an order from lower to higher ratings.
City	Categorical	Nominal	City names are categories and do not have a natural order.
Open/Closed Status	Categorical	Nominal	Open and Closed are categories without any ranking.

Conclusion:
Restaurant Name, Cuisine Type, City, and Open/Closed Status are categorical variables. Average Cost for Two and User Rating are numerical variables, with User Rating also representing an ordered scale.

Task 2

Question:
Create a table with 8 real-life examples of data you might find in popular apps like Instagram, Flipkart, or IRCTC. For each, specify the measurement scale (Nominal, Ordinal, Interval, or Ratio) and briefly explain your choice.

Answer:

No.	Real-Life Example	Measurement Scale	Explanation
1	Instagram – Number of Likes	Ratio	Likes have a true zero, and 20 likes are twice as many as 10 likes.
2	Flipkart – Product Category	Nominal	Categories such as Electronics, Clothing, and Furniture have no natural order.
3	Flipkart – Product Rating	Ordinal	Ratings from 1 star to 5 stars have a meaningful order.
4	IRCTC – Train Departure Time	Interval	The difference between times is meaningful, but 12:00 midnight does not represent an absence of time.
5	Flipkart – Number of Products Ordered	Ratio	Zero means no products were ordered, and ratios are meaningful.
6	Instagram – Account Type	Nominal	Personal, Business, and Creator are categories without a ranking.
7	IRCTC – Waiting List Position	Ordinal	Positions such as 1, 2, and 3 represent an order or ranking.
8	Flipkart – Delivery Distance	Ratio	Distance has a true zero and ratios are meaningful.

Conclusion:
Nominal data represents categories, ordinal data represents ordered categories, interval data has equal intervals without a true zero, and ratio data has equal intervals with a true zero.

Task 3

Question:
Given this time-series data from a Spotify user's listening history — Date, Song Name, Play Count per Day — identify which columns represent time-series, and which are categorical or numerical. Explain your reasoning.

Answer:

Column	Data Type	Classification	Explanation
Date	Time-Series / Temporal	Time-Series	It records listening activity according to different dates.
Song Name	Categorical	Nominal	Song names are labels and do not have a natural order.
Play Count per Day	Numerical – Discrete	Ratio	It represents the number of times a song was played. Zero means it was not played.

Example:

Date	Song Name	Play Count per Day
01-09-2026	Song A	5
02-09-2026	Song B	3
03-09-2026	Song A	8

Explanation:
The **Date** column represents time-series data because the observations are recorded over time. **Song Name** is categorical because it identifies different songs. **Play Count per Day** is numerical and discrete because it represents a count.

Task 4

Question:
Write a function or pseudocode that takes a column of data (array of values) from a Flipkart order history and automatically detects whether it is likely Nominal, Ordinal, Interval, or Ratio scale. Briefly describe the logic used for each type.

Answer:

Pseudocode

```
FUNCTION detect_scale(column):

    IF column contains categories or names:
        RETURN "Nominal"

    ELSE IF column contains ordered categories:
        RETURN "Ordinal"

    ELSE IF column contains dates, years, or temperature values:
        RETURN "Interval"

    ELSE IF column contains numerical values
            with a meaningful zero:
        RETURN "Ratio"

    ELSE:
        RETURN "Unknown"
```

Explanation

1. Nominal Scale

Used when values are **names or categories with no natural order**.

Example:

```
Payment Method:
UPI
Cash
Credit Card
```

Result:

```
Nominal
```

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2. Ordinal Scale

Used when values have a **meaningful order or ranking**.

Example:

```
Delivery Rating:
Poor
Average
Good
Excellent
```

Result:

```
Ordinal
```

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3. Interval Scale

Used when the **difference between values is meaningful**, but there is no true zero.

Example:

```
Temperature:
10°C
20°C
30°C
```

Result:

```
Interval
```

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4. Ratio Scale

Used when the data has a **true zero** and ratios are meaningful.

Example:

```
Product Price:
₹0
₹500
₹1000
```

Result:

```
Ratio
```